Directorate of Artificial Intelligence & Visualisation,
Research Centre Imarat (RCI), Defence Research & Development Organisation (DRDO)
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LITERATURE SURVEY

ON

CROSS-SPECTRAL DSMAC

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INTRODUCTION

What is DSMAC?

- DSMAC (Digital Scene Matching Area Correlation) is a vision-based navigation technique that estimates a vehicle's position by matching onboard sensor imagery with reference maps.
- It is widely used in GPS-denied environments where satellite navigation is unavailable or unreliable.
- DSMAC compares observed scene features with stored terrain or scene information to determine accurate localization.

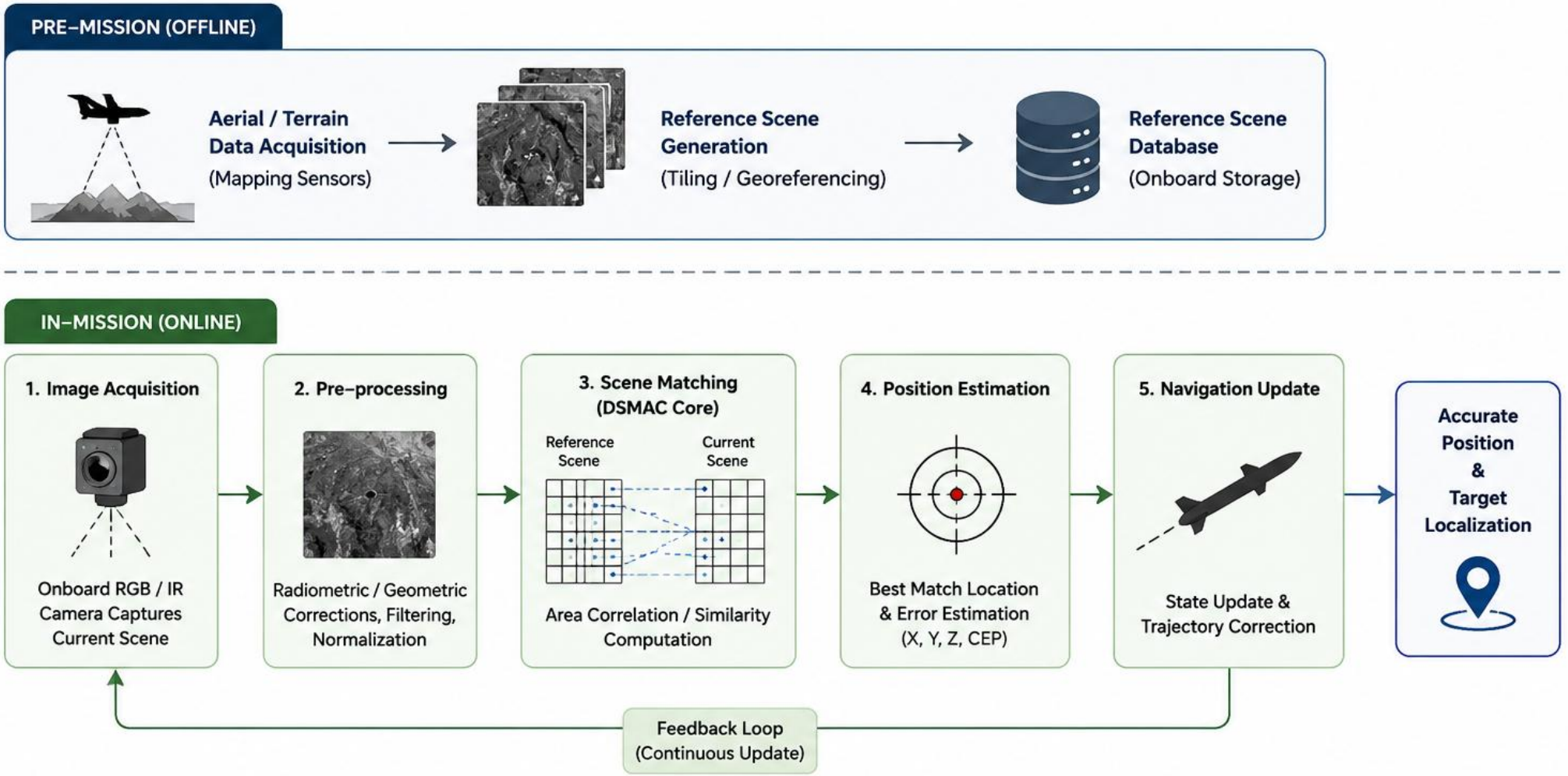
Why is DSMAC Important?

- Provides autonomous navigation without dependence on GPS.
- Enhances robustness in contested or signal-denied environments.
- Applicable to aerial vehicles, missiles, autonomous robots, and unmanned systems.
- Enables precise localization using onboard vision sensors.

Motivation

Modern navigation systems increasingly employ multiple sensing modalities, such as RGB and SAR (Synthetic Aperture Radar) seeker, to improve robustness under varying environmental and illumination conditions.

DSMAC – GENERAL WORKING PIPELINE



EVOLUTION OF DSMAC

1980

Original DSMAC [1]

Approach

- Area-based image correlation
- Correlation Surface Peak Detection
- Pixel-wise scene comparison
- DSMAC-assisted terminal guidance
- Reference image matching

Limitations

- Sensitive to illumination and viewpoint changes
- High computational complexity
- Limited adaptability

1994

Advanced Scene Matching [2]

Approach

- Enhanced image processing techniques
- Improved scene registration
- Robust geometric alignment
- Optimized image matching for terminal guidance

Limitations

- Still relied on handcrafted image processing
- No learned feature representation
- Sensitive to environmental variations

2026

AI-Based DSMAC [3]

Approach

- Deep feature representation
- ResNet-50 feature learning
- FAISS similarity search
- CNN-based aerial image matching
- Automated dataset generation

Limitations

- High computational requirements
- Limited multimodal (RGB–IR) capability

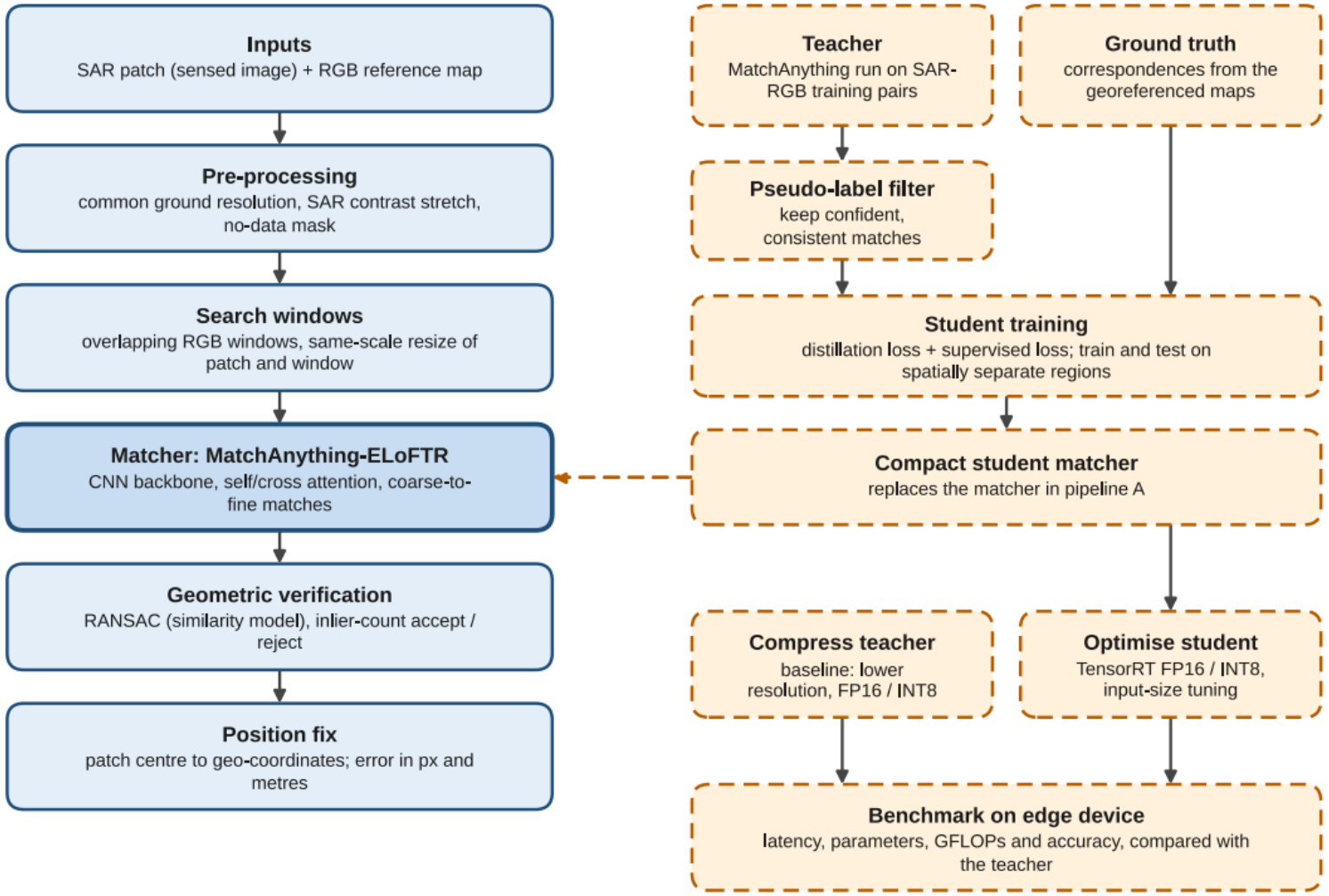
Proposed Work

Lightweight Cross-Modal (RGB–SAR) Scene Matching for Edge Deployment

LITERATURE REVIEW

Method / Paper	Year	Matching Type	Core Approach	Main Contribution	Reported Accuracy (dataset, metric)	Latency (hardware)	Relevance to SAR–RGB DSMAC
LoFTR	2021	Dense image matching	Transformer-based coarse-to-fine matching without explicit keypoint detection	Enables dense correspondence using global context	SpaceNet9 SAR–optical: mean error 5.1 px; success@5 px 54.4%, success@10 px 81.0%	66.2 ms (MegaDepth, 1200×1200, RTX 3090)	Important architectural background for modern learned matching; not trained for SAR
EfficientLoFTR	2024	Efficient dense matching	Aggregated attention with adaptive token selection and two-stage correlation refinement	Reduces computational cost while retaining dense matching capability	No SAR–optical result in the paper. MegaDepth pose AUC@5°: 56.4 (LoFTR 52.8)	40.1 ms FP32, 34.4 ms mixed precision (MegaDepth, 1200×1200, RTX 3090)	Relevant to eventual onboard/real-time implementation; the matcher inside MatchAnything-ELoFTR
XoFTR	2024	Cross-modal dense matching (visible–thermal)	LoFTR-style transformer pre-trained with masked image modelling, fine-tuned with pseudo-thermal augmentation	Cross-modal matching despite large appearance change	SpaceNet9 SAR–optical: mean error 3.0 px; success@5 px 78.4%, success@10 px 90.5%	About 0.4 s per image pair under the tiled protocol (RTX 3090)	Among the best SAR–optical results in (tied with RoMa); trained on visible–thermal data, not SAR
Chen et al.	2023	SAR–optical (sparse detect-and-describe)	Shared-weight CNN with a co-attention matching module that builds each image's keypoint map from both images	Pair-wise keypoint detection; positive-sample pooling and negative-sample augmentation for the descriptor	SEN1-2 test pairs: repeatability 0.434, NN mAP 0.309, average corner error 7.1 px (SIFT and SAR-SIFT failed; RIFT 49.8; SuperPoint 37.0; R2D2 27.0)	Not reported (trained on an RTX 3060)	Independent evidence on SAR–optical pairs; a small CNN suits edge use, but latency is unreported
MatchAnything	2025	General cross-modal	Large-scale cross-modal pretraining of existing matchers (ELoFTR, RoMa)	Learns modality-general matching using synthetic cross-modal training signals and demonstrates transfer to unseen modalities	SpaceNet9 SAR–optical [A], ELoFTR variant: mean error 3.4 px; success@5 px 71.2%, success@10 px 93.3%	ELoFTR architecture: 40.1 ms (1200×1200, RTX 3090) ; not re-measured for SAR	Very relevant conceptually because it addresses generalisation across unseen modalities; SAR–RGB applicability must be verified experimentally
DistillMatch	2025	Multimodal	DINOv2/v3 → lightweight student via knowledge distillation	Efficient multimodal matching	METU-VisTIR: AUC@5° 23.13/12.45, AUC@10° 41.10/26.86, AUC@20° 58.41/44.47 (cloud–cloud / cloud–sunny). Zero-shot optical–SAR: evaluated using NCM (<5 px) and RMSE.	Inference latency: Not reported. Training: 120 h on 3× NVIDIA RTX 4090, 20 epochs, batch size 1.	Relevant to lightweight SAR–RGB adaptation

APPROACH


Inputs
SAR patch (sensed image) + RGB reference map
Pre-processing
common ground resolution, SAR contrast stretch, no-data mask
Search windows
overlapping RGB windows, same-scale resize of patch and window
Matcher: MatchAnything-ELoFTR
CNN backbone, self/cross attention, coarse-to-fine matches
Geometric verification
RANSAC (similarity model), inlier-count accept / reject
Position fix
patch centre to geo-coordinates; error in px and metres
Teacher
MatchAnything run on SAR-RGB training pairs
Ground truth
correspondences from the georeferenced maps
Pseudo-label filter
keep confident, consistent matches
Student training
distillation loss + supervised loss; train and test on spatially separate regions
Compact student matcher
replaces the matcher in pipeline A
Compress teacher
baseline: lower resolution, FP16 / INT8
Optimise student
TensorRT FP16 / INT8, input-size tuning
Benchmark on edge device
latency, parameters, GFLOPs and accuracy, compared with the teacher

IMPLEMENTATION

Steps Implemented:

1. Dataset & Geospatial Preparation

- ICEYE SAR + Google RGB imagery
- Common geographic coverage
- SAR resampled to RGB grid
- CRS/georeferencing verified
- SAR no-data masking + percentile normalization

2. Patch Generation

- Geographically co-located SAR–RGB patches
- Controlled patch size: 2048×2048 for MatchAnything experiments
- Smaller patches used during student experiments
- Training/holdout spatial split

3. Baseline Matcher MatchAnything–ELoFTR

- SAR–RGB cross-modal correspondence extraction
- Sliding RGB search windows
- Similarity/affine geometric verification using RANSAC

4. Geolocation Evaluation

- Predicted patch centre → RGB pixel coordinates
- RGB pixel coordinates → latitude/longitude
- Geodesic localization error in metres
- Metrics: matches, inliers, pixel error, localization error

Steps Next to be Implemented:

5. RGB-SAR Student Model Development

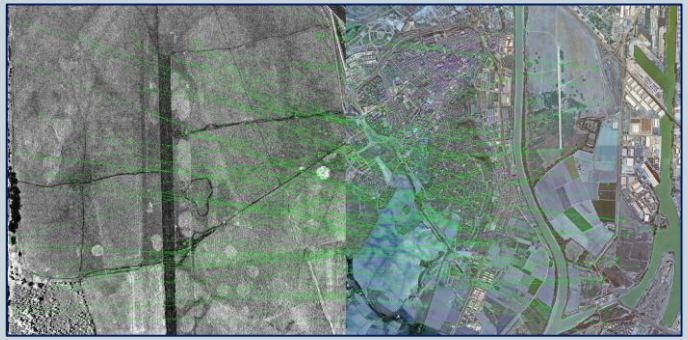
6. Distillation Training

7. Edge Optimisation

PRELIMINARY RESULTS

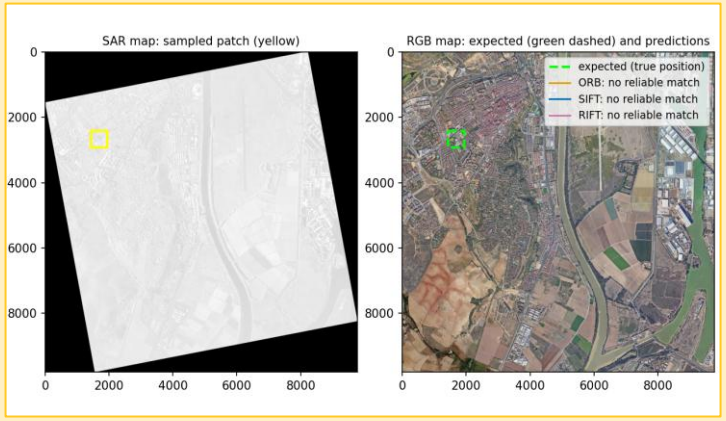
ELoFTR

- Irrelevant Scene Matching
  No reliable match found
- Reason: ELoFTR not inherently desgned for cross modality

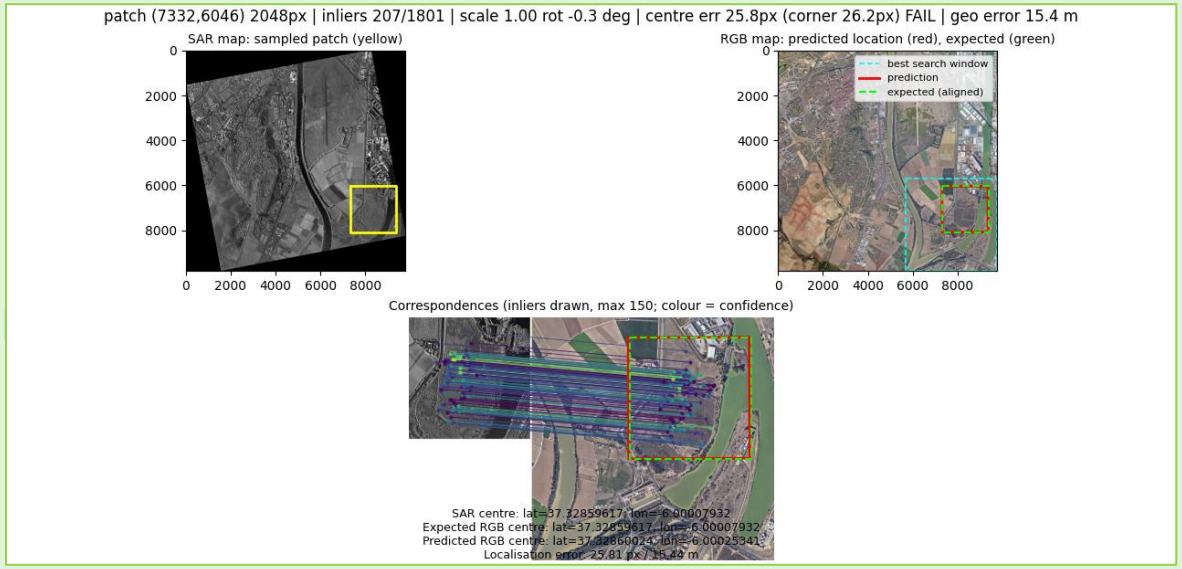


SIFT, ORB, RIFT

- No reliable match found in any method
- Reason: handcrafted local features lack modality-invariant descriptors.



MatchAnything-ELoFTR



SAR centre: lat=37.32859617;lon =- 6.00007932
Expected RGB centre: lat=37.32859617.lon=6.00007932
Predicted RGB centre: lat=37.32860024. lon =- 6.00025341
Localisation error 25.81 px / 15.44 m
Threshold: 10m

SUMMARY & CONCLUSION

Summary

- Cross-modal image matching is an important component for robust DSMAC operation in GPS-denied environments.
- The literature shows significant progress in learned and cross-modal matching, but **SAR–RGB matching remains challenging due to the large differences in imaging characteristics between the two modalities**.
- Modern methods such as MatchAnything demonstrate strong cross-modal matching capability, while DistillMatch shows the potential of **knowledge distillation for developing lightweight multimodal matching models**.
- Our preliminary implementation establishes a **SAR–RGB matching and geolocation evaluation pipeline**, with MatchAnything–ELoFTR achieving a localization error of **15.44 m** on the tested scene.
- These results motivate the development of a **dedicated lightweight SAR–RGB matcher** rather than directly relying on a general-purpose pretrained matcher.

Conclusion

- The proposed direction is to develop a **lightweight cross-modal student model**, trained using geospatially aligned SAR–RGB data and knowledge transferred from a strong teacher matcher.
- The framework will focus on learning **cross-modal correspondence and geometric information**, while retaining modality-specific characteristics important for SAR–RGB matching.
- The next stages are **student-model development, distillation training, edge optimization, and quantitative evaluation** in terms of localization accuracy, matching reliability, computational cost, and inference latency.
- The overall objective is to achieve a practical **accuracy–efficiency trade-off suitable for edge/onboard deployment**.

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